Distributed Ledger Technology & Blockchain:

Potential implications on the financial sector

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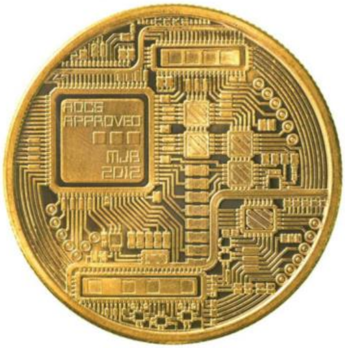
What is a Cryptocurrency?

(Also referred to as Digital currencies)



Cryptocurrencies are digital representations of value that are denominated in their own unit of account, distinct from e-money.

Distinct characteristics:

1) Not backed by an underlying asset (have no intrinsic value; value is determined by demand & supply)
2) Do not represent a liability on any institution
3) Exchanged through **distributed ledgers absent central record keeping** and absent trust between network members
4) Do not rely on institutional arrangements or intermediaries for P2P transfers

Motivations:

- Solve "double-spend" problem
- Eliminate need for trusted central party & enable P2P transfers of digital value
- Eliminate government control over money supply

Risks:

- High price volatility driven by speculative behavior
- Lack of regulation/uncertainty about regulation
- Data loss problem
- Not widely accepted as unit of exchange for goods & services; conversion to fiat currency necessary

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Source: CPMI, "Digital Currencies", BIS, November 2015

How do Cryptocurrencies work?

- Essentially a set of independent entities [nodes] running a particular software [open source] each maintaining records of transactions. *[Blockchain - Distributed Ledger] - nodes are participating out of free-will.*

- A particular base amount of CC created at inception and subsequent ones created as a reward to nodes facilitating propagation of new transactions – no entity backing CC and it not guaranteed by anybody.

- The propagation of new transactions by design requires significant computing power – to impose a cost and deter malicious attacks; new CCs created as a reward. *[Consensus process]*

- Others can buy CC from existing owners or from online exchanges [akin to money changers]
- Ownership of CC tied to a key pair – public key [like email id] and private key [like email password]

- Transactions recorded in the blockchain, each Txn tied to a public key

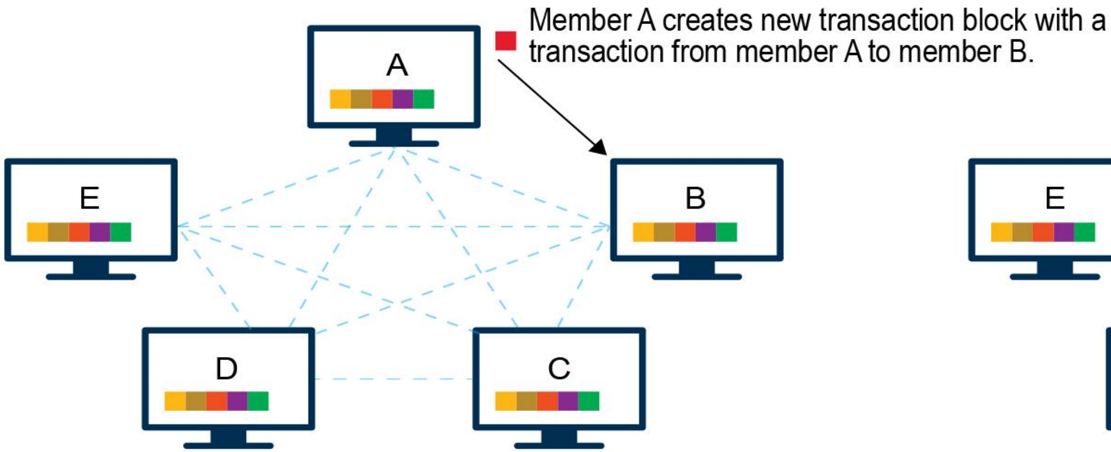
What is the difference between DLT and Blockchain?

- Blockchain underlying technology used in Bitcoin and many other CCs.

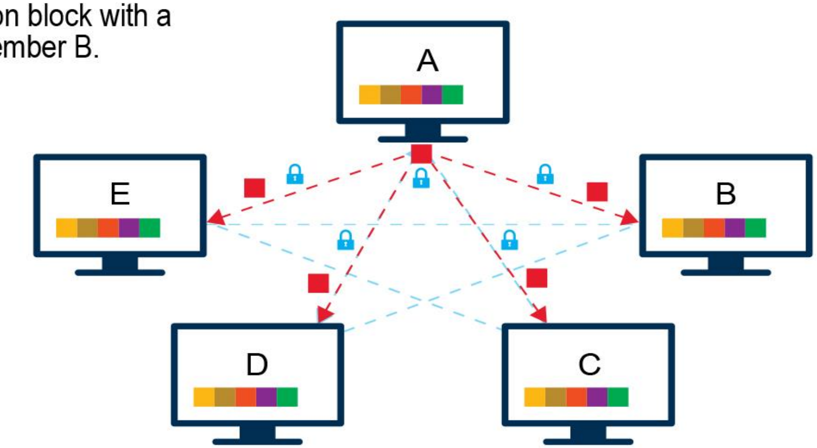
- Distributed Ledger Technology is the broad class of technology and Blockchain is one variant of it.

- Distributed Ledgers can record ownership and transactions of any “digital asset”:
  - Cryptocurrencies
  - Central Bank issue Digital Currency
  - Record of land-ownership [land identified by a code]
  - Record of ownership of a particular lot of farm produce
  - Origin certificate for a agricultural product or commodity
  - Identity information

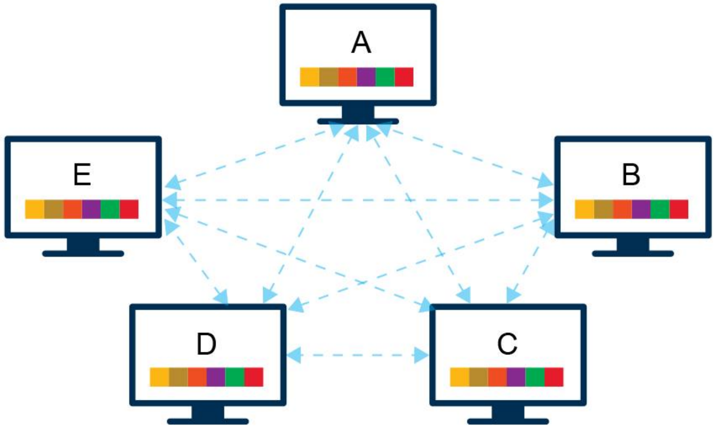
But….. How does it work?



1) New additions to the database are initiated by one of the members (nodes), who creates a new “block” of data containing several transaction records.



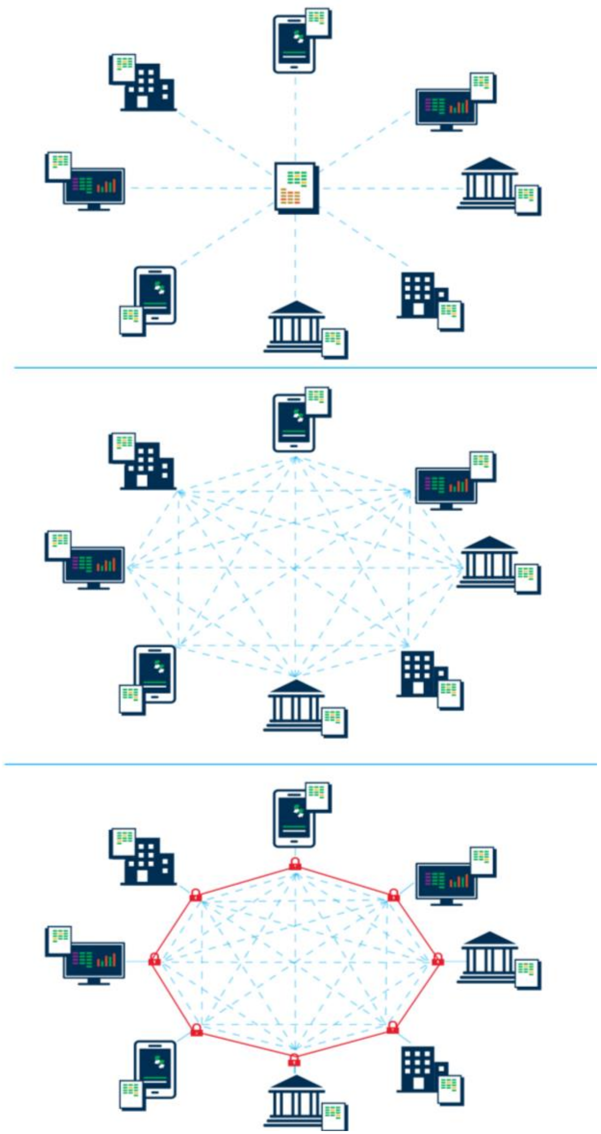
2) Information about this new data block is then shared across the entire network, containing encrypted data so transaction details are not made public.



3) All network participants collectively determine the block’s validity according to a pre-defined **algorithmic validation method (‘consensus mechanism’).** Only after validation, all participants add the new block to their respective ledgers.

Source: Adapted from WSJ graphic in “Dubai Aims to Be a City Built on Blockchain”, 24 April 2017 in: Distributed Ledger Technology and Blockchain, Fintech Note #1

What is a Distributed Ledger?



Centralized Ledger
All parties reconcile their local databases with a centralized electronic ledger that is maintained and controlled by a trusted central party.

Distributed Ledger (permissionless)
In a permissionless distributed ledger, all network members (nodes) collectively validate proposed additions to the ledger through an algorithmic consensus mechanism. Each node can propose additions to the ledger and owns a full and up-to-date copy of the entire ledger.

Distributed Ledger (permissioned)
In a permissioned system, nodes need permission from a central entity to access the network and make changes to the ledger. Access controls can include identity verification.

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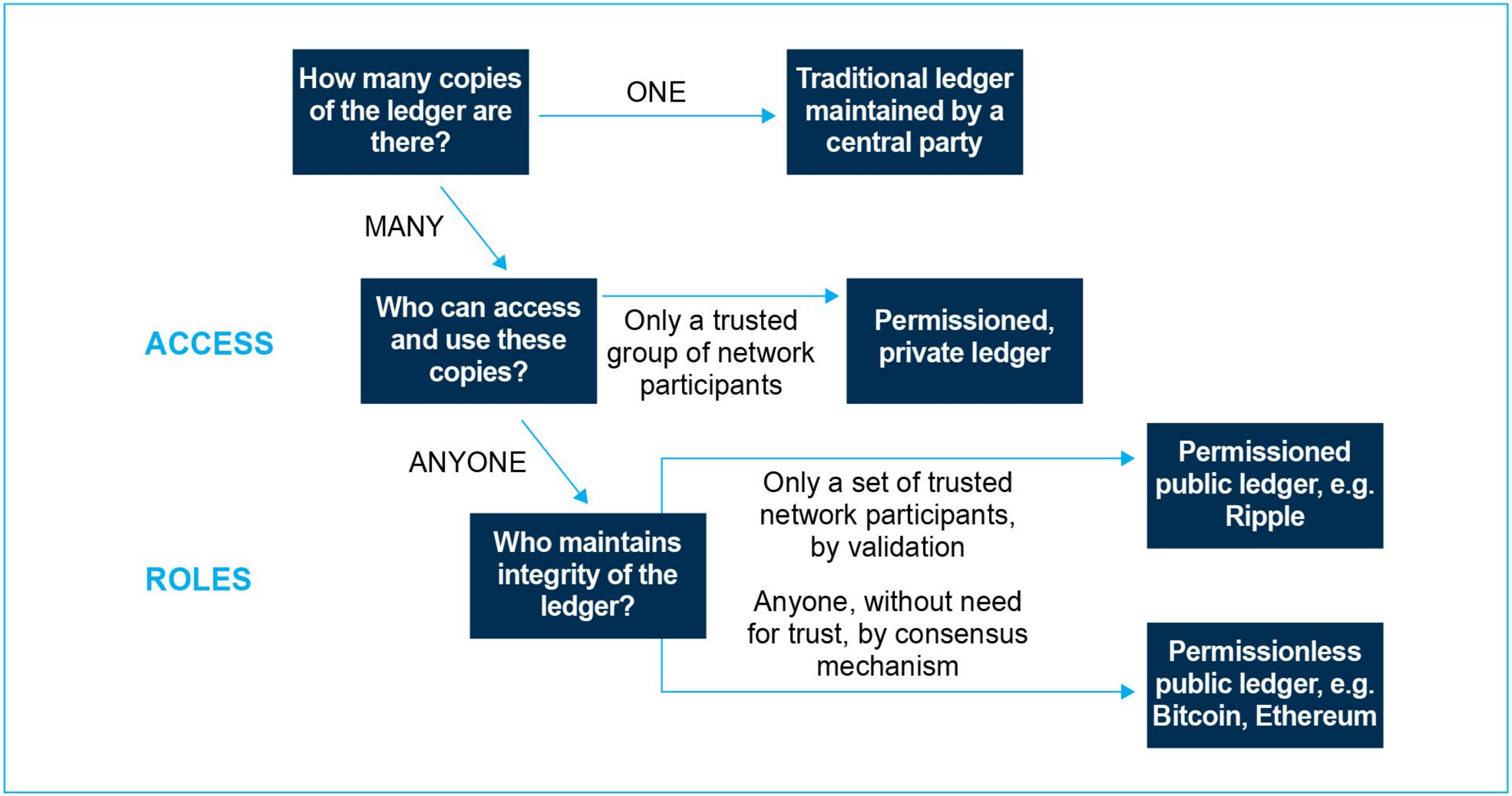
Source: Distributed Ledger Technology and Blockchain, Fintech Note #1

Summing up….. Distributed Ledger Taxonomy



WORLD BANK GROUP *Source: Adapted from Dave Birch (Consult Hyperion) in: Distributed Ledger Technology and Blockchain, Fintech Note #1*

You name it! As many implications as you want…

Payments and financial markets

1

- Cross Border Payments
- Micro Payments
- Depositories, Stock Exchanges
- Corporate Actions, Derivative Contracts
- Digital Identity

Lending, investment and insurance

2

- Syndicated loans
- Lending platforms, Crowd-funding and ICOs
- Insurance contracts as Smartcontracts
- Trade Finance

Asset Registries and Records

3

- Collateral registries
- Record of provenance of agricultural products and commodities

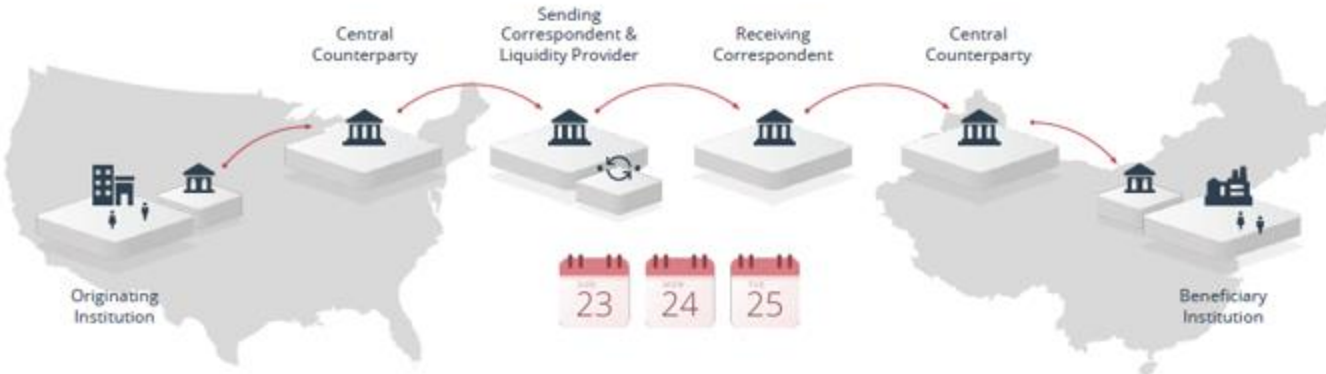
Internal Systems of financial institutions

4

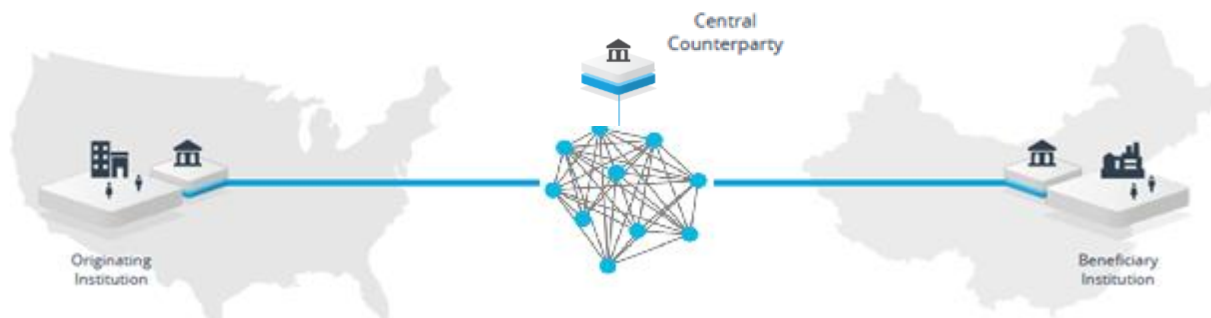
- Internal systems of multinational financial institutions
- Even between same organization

1 Clearly on cross-border payments…

Today: SWIFT + Correspondent = Relay



Tomorrow 1.0: trusted protocol – Bank2Bank

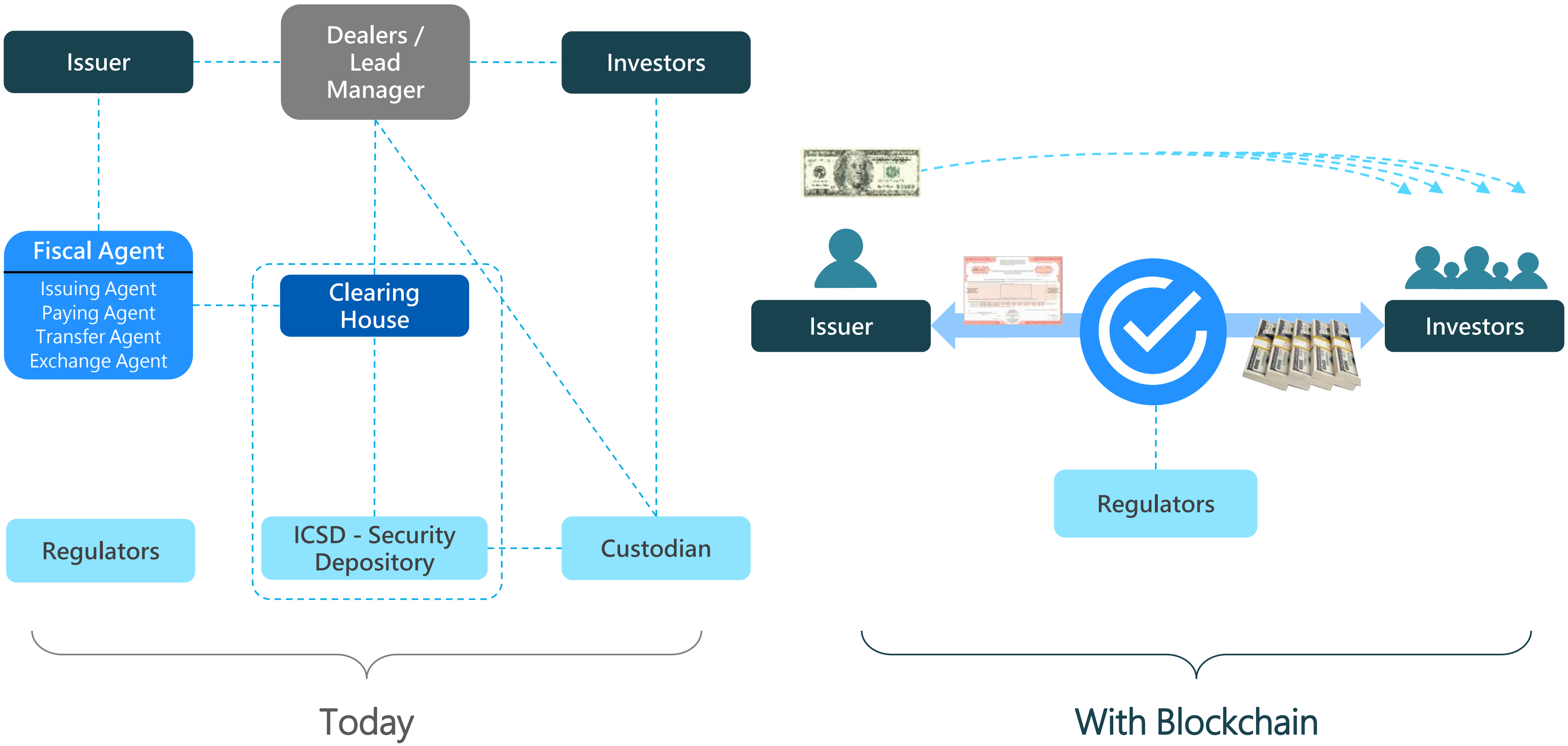


Tomorrow 2.0 : party to party = No Banks?

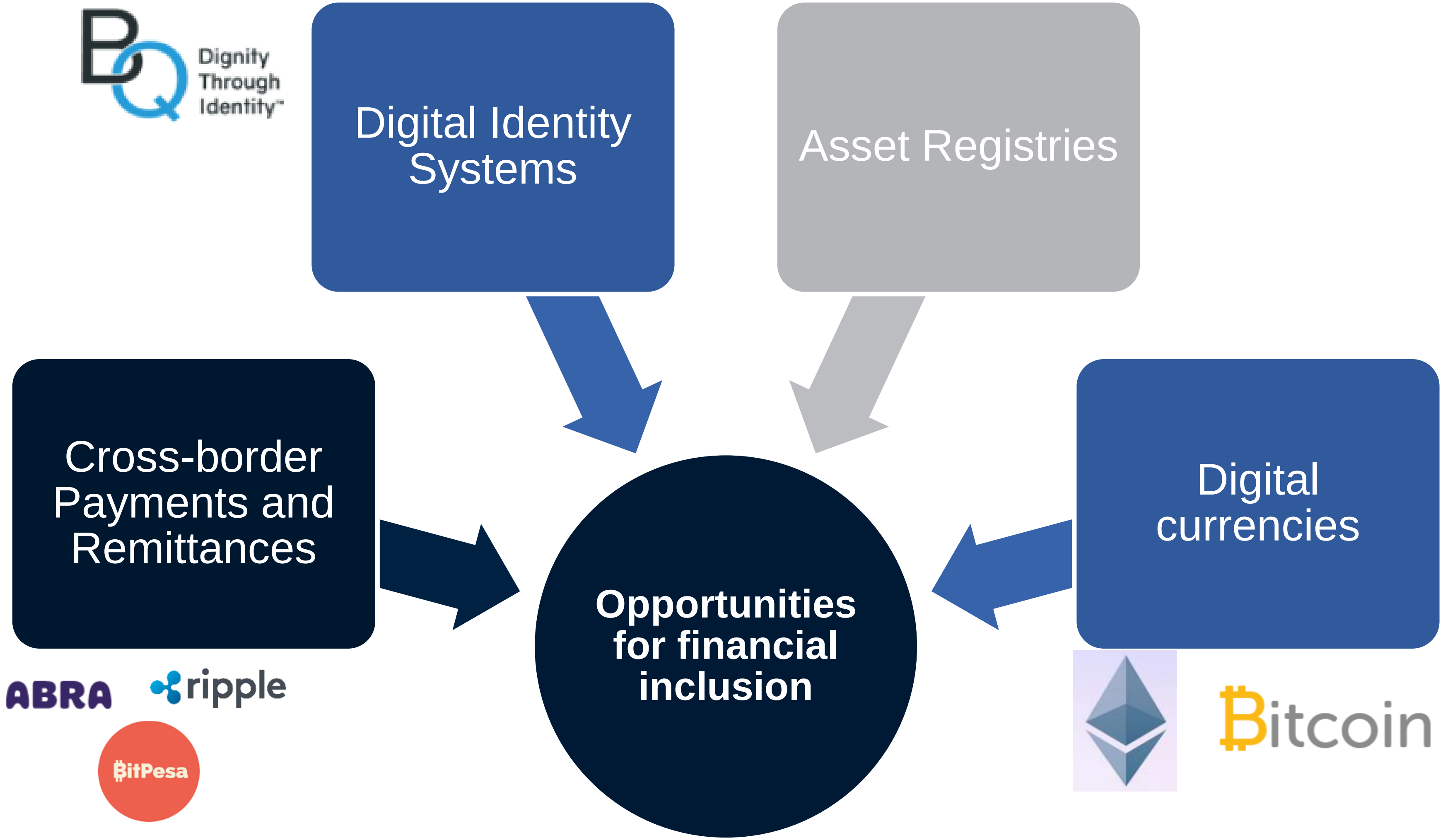
2 Also on Trade Finance…

WORLD BANK GROUP *Source: ING/Wall Street Journal*

3 And on capital markets (bond issuance)…



4 And potentially on financial inclusion through many different channels…



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Challenges and risks…
So many questions to be answered

- High environmental costs from mining
- Not immune to cyber attacks or hacks
- Verification speed and transaction volumes (Scalability?)
- Lack of technological maturity
- Unclear governance or enforcement
- ?
- Regulatory vetting and industry standards
- Privacy issues
- Unclear recourse mechanisms
- Unclear legal ownership and jurisdiction
- KYC & CDD requirements
- Interoperability and integration in financial sector

WB Position on Digital Currencies

- The WB should encourage all client countries to develop an appropriate **regulatory framework**, warn the public on the risks and monitor any build-up of linkages of their financial system with the DC ecosystem.

- The DCs pose several risks and **at this point are unsuitable for advancing financial inclusion**. In the medium term, public authorities should continue the focus on existing approaches to promote financial inclusion.

- The applications of DCs, CBDCs and eDCs **for cross-border payments and regional payment platforms hold promise**. The World Bank should actively contribute to the testing and help shape the various solutions being piloted by private sector players.

- There is a need for **international co-operation and co-ordination** on DCs and CBDCs. The FATF, FSB, G20 and GPFI mechanism could be leveraged for this.

- DLT - the underlying technology of DC have wide applications in the financial sector and hold more promise.

Potential World Bank Actions

Monitor developments:

- Leverage IFC investees and forums for knowledge exchange and identify feasible design and implementation pilots

Foster collaboration and coordination:

- Join industry consortiums
- Foster international co-operation and collaboration

Enhance Awareness & Explore:

- Encourage exploration of applicability of DLT and conduction of pilots
- Enhance level of awareness on DLT
- Establish a blockchain lab
- Encourage country counterparts to include DLT solutions
- Explore financing small-scale pilots

Actively Engage with WB Clients:

- Support WB countries in establishing sandboxes, hubs, accelerators or pilots

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Thank you!

Questions?